```
praveen@PRAVEENKUMAR:~$ mkdir -p ~/process_practice
praveen@PRAVEENKUMAR:~$ cd ~/process_practice
praveen@PRAVEENKUMAR:~/process_practice$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.5 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [803 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [184 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [374 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [118 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 2305 kB in 2s (1078 kB/s)
31 packages can be upgraded. Run 'apt list --upgradable' to see them.
praveen@PRAVEENKUMAR:~/process_practice$ sudo apt install -y procps psmisc htop
procps is already the newest version (2:4.0.4-9ubuntu1).
procps set to manually installed.
psmisc is already the newest version (23.7-2ubuntu2).
psmisc set to manually installed.
Installing:
  htop

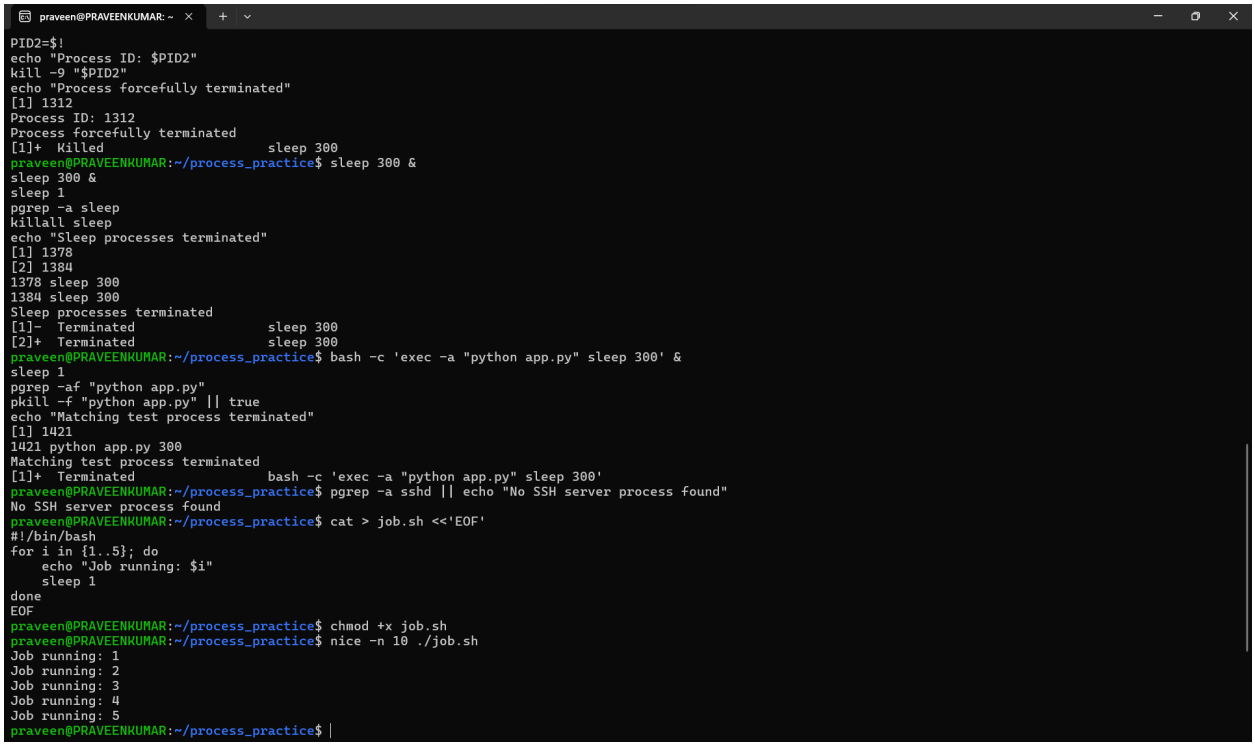
Suggested packages:
  lm-sensors  strace

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 31
  Download size: 177 kB
  Space needed: 451 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 2s (103 kB/s)
Selecting previously unselected package htop.
(Reading database ... 35907 files and directories currently installed.)
Preparing to unpack .../htop_3.4.1-5build2_amd64.deb ...
Unpacking htop (3.4.1-5build2) ...
Setting up htop (3.4.1-5build2) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
praveen@PRAVEENKUMAR:~/process_practice$ ps
    PID TTY          TIME CMD
    395 pts/0    00:00:00 bash
```

```
praveen@PRAVEENKUMAR:~/process_practice$ ps
    PID TTY          TIME CMD
    395 pts/0    00:00:00 bash
   1215 pts/0    00:00:00 ps
praveen@PRAVEENKUMAR:~/process_practice$ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24536 15532 ?        Ss   09:30   0:00 /sbin/init
root           2  0.0  0.0   3180  2212 hvc0     Sl+  09:30   0:00 /init
root           8  0.0  0.0   3180  2040 hvc0     Sl+  09:30   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          48  0.0  0.2  50400 16868 ?        S<s  09:30   0:00 /usr/lib/systemd/systemd-journald
systemd+      80  0.0  0.1  22432 14428 ?        Ss   09:30   0:00 /usr/lib/systemd/systemd-resolved
root          86  0.2  0.1  35424 12368 ?        Ss   09:30   0:01 /usr/lib/systemd/systemd-udevd
root         106  0.0  0.0   2888  2012 ?        Ss   09:30   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         107  0.0  0.0   4472  3140 ?        Ss   09:30   0:00 /usr/sbin/cron -f -P
message+     108  0.0  0.0   8828  5412 ?        Ss   09:30   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
og-only
praveen@PRAVEENKUMAR:~/process_practice$ top -b -n 1 | head -15
top - 09:45:53 up 15 min,  1 user,  load average: 0.09, 0.05, 0.01
Tasks:  29 total,   1 running,  28 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.4 id,  0.6 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7783.6 total,   7187.2 free,    568.0 used,    182.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7215.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24536  15532  11628 S   0.0   0.2   0:00.66 systemd
      2 root      20   0    3180   2212   2072 S   0.0   0.0   0:00.01 init-sy+
      8 root      20   0    3180   2040   1940 S   0.0   0.0   0:00.00 init
     48 root      19  -1   50400  16868  15572 S   0.0   0.2   0:00.21 systemd+
     80 systemd+  20   0   22432  14428  11936 S   0.0   0.2   0:00.07 systemd+
     86 root      20   0   35424  12368   9216 S   0.0   0.2   0:01.96 systemd+
    106 root      20   0    2888   2012   1884 S   0.0   0.0   0:00.04 chronyd+
    107 root      20   0    4472   3140   2892 S   0.0   0.0   0:00.00 cron
praveen@PRAVEENKUMAR:~/process_practice$ htop -b -n 1 | head -15
htop: invalid option -- 'b'
praveen@PRAVEENKUMAR:~/process_practice$ sleep 300 &
PID1=$!
echo "Process ID: $PID1"
kill "$PID1"
echo "Process terminated"
[1] 1278
Process ID: 1278
Process terminated
[1]+  Terminated                sleep 300
praveen@PRAVEENKUMAR:~/process_practice$ sleep 300 &
PID2=$!
echo "Process ID: $PID2"
kill -9 "$PID2"
```


praveen@PRAVEENKUMAR: ~
PID2=$!
echo "Process ID: $PID2"
kill -9 "$PID2"
echo "Process forcefully terminated"
[1] 1312
Process ID: 1312
Process forcefully terminated
[1]+ Killed   sleep 300
praveen@PRAVEENKUMAR:~/process_practice$ sleep 300 &
sleep 300 &
sleep 1
pgrep -a sleep
killall sleep
echo "Sleep processes terminated"
[1] 1378
[2] 1384
1378 sleep 300
1384 sleep 300
Sleep processes terminated
[1]- Terminated   sleep 300
[2]+ Terminated   sleep 300
praveen@PRAVEENKUMAR:~/process_practice$ bash -c 'exec -a "python app.py" sleep 300' &
sleep 1
pgrep -af "python app.py"
pkill -f "python app.py" || true
echo "Matching test process terminated"
[1] 1421
1421 python app.py 300
Matching test process terminated
[1]+ Terminated   bash -c 'exec -a "python app.py" sleep 300'
praveen@PRAVEENKUMAR:~/process_practice$ pgrep -a sshd || echo "No SSH server process found"
No SSH server process found
praveen@PRAVEENKUMAR:~/process_practice$ cat > job.sh <<'EOF'
#!/bin/bash
for i in {1..5}; do
    echo "Job running: $i"
    sleep 1
done
EOF
praveen@PRAVEENKUMAR:~/process_practice$ chmod +x job.sh
praveen@PRAVEENKUMAR:~/process_practice$ nice -n 10 ./job.sh
Job running: 1
Job running: 2
Job running: 3
Job running: 4
Job running: 5
praveen@PRAVEENKUMAR:~/process_practice$